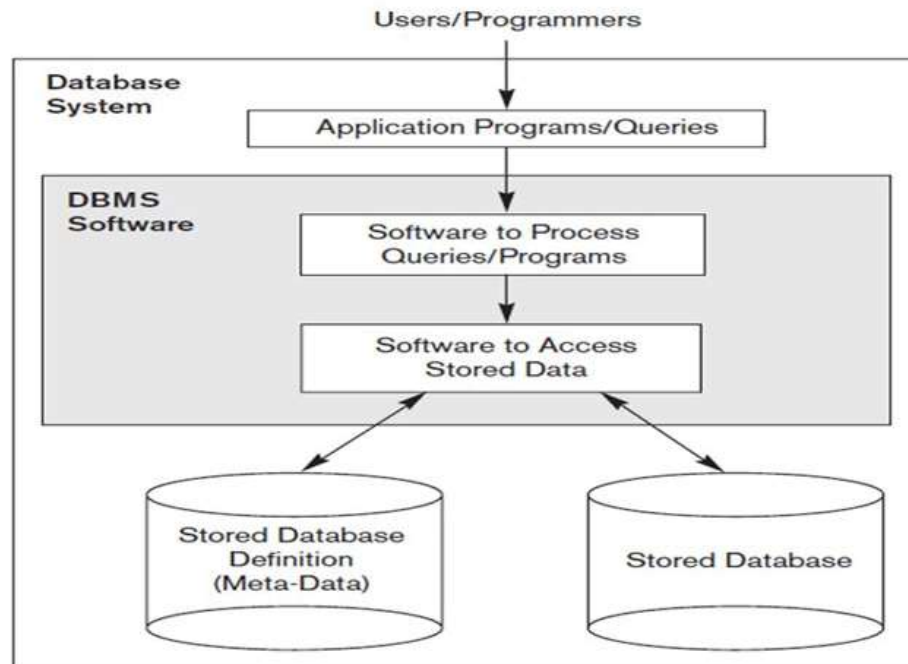


❖ Sketch and explain a simplified database environment.

Simplified Database System Environment



❖ Distinguish among SQL, MySQL, Oracle and SQL server.

Feature	SQL	MySQL	Oracle	SQL Server
Developer	IBM	Oracle	Oracle	Microsoft
License	Open source	Open source	Proprietary	Proprietary
Primary Language	ANSI SQL	ANSI SQL	SQL and PL/SQL	T-SQL
Platform	Cross-platform	Cross-platform	Cross-platform	Windows only

❖ **Difference between CHAR and VARCHAR in database using table format.**

Feature	CHAR	VARCHAR
Length	Fixed length	Variable length
Storage Size	Length x Character size	Length + Actual data size
Efficiency	Fixed length increases efficiency	Variable length can be less efficient
Usage	For storing fixed-length data	For storing variable-length data
Retrieval speed	Retrieval speed is faster	Retrieval speed is slower

❖ **Explain different types of SQL constraints and integrity constraints.**

There are different types of SQL constraints:

1. **NOT NULL Constraint:** This constraint ensures that a column does not contain NULL values.
2. **UNIQUE Constraint:** This constraint ensures that a column or a combination of columns have unique values in the table.
3. **PRIMARY KEY Constraint:** This constraint is a combination of NOT NULL and UNIQUE constraints.
4. **FOREIGN KEY Constraint:** This constraint ensures referential integrity between two tables.

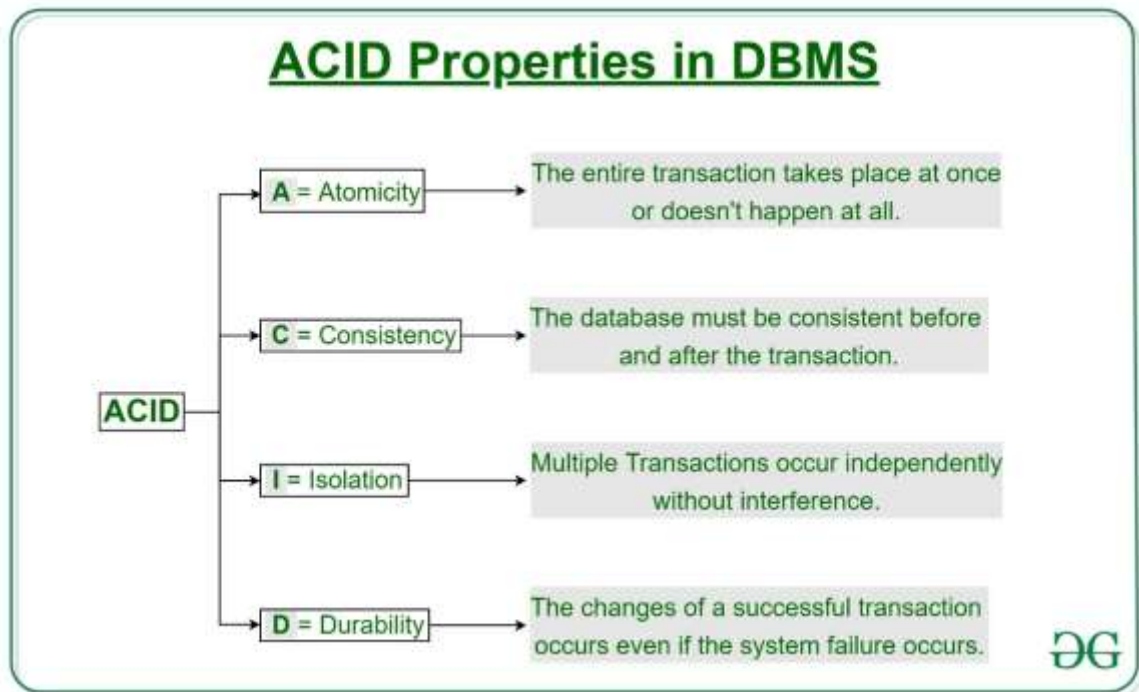
There are different types of integrity constraints:

1. **Entity Integrity Constraint:** This constraint ensures that every row in a table is uniquely identified by a primary key.
2. **Referential Integrity Constraint:** This constraint ensures that the relationships between tables are consistent.
3. **Domain Integrity Constraint:** This constraint ensures that the data entered into a column of a table is within a specific domain or range of values
4. **User-Defined Integrity Constraint:** This constraint is a custom constraint that is created by the user to ensure that specific business rules are enforced in the database.

❖ What is query processor?

A query processor, also known as a query optimizer, is a component of a database management system (DBMS) that is responsible for optimizing queries and generating an execution plan to retrieve data from the database.

❖ ACID properties in DBMS.



❖ Discuss two reasons why null values might be introduced into the database.

- **Missing Data:** Null values can be introduced into the database when data is missing or unknown. For example, if a customer has not provided their phone number or email address, these fields may be left blank, resulting in null values.
- **Data Transformations:** Null values can also be introduced into the database as a result of data transformations or operations. For example, if data is imported from a file and a required field is missing from the file, the database may insert a null value into the corresponding column.